$\wedge$ AND $\vee$ OR $\neg$ NOT

COMP 2121 DISCRETE MATHEMATICS — LAB 3

$a = 1$; $b \wedge c = 0$

x	y	$x \to y$
0	0	1
0	1	1
1	0	0
1	1	1

$x \to y \Leftrightarrow \neg x \vee y$ No NAME

$x \to y \Leftrightarrow \neg y \to \neg x$ Contrapositive

1. Let $\boxed{a \to (b \wedge c)}$ be false. What is the truth value of the following statements:

   No NAME

   i) $a \wedge b \wedge c$

   $1 \wedge 0$

   $= 1$

   ii) $\boxed{\neg a \vee (b \wedge c)}$

   $\neg 1 \vee 0$

   $= 0 \vee 0$

   $= 0$

   Contrapositive

   iii) $(b \wedge c) \to a$

   $0 \to 1$

   $= 1$

   iv) $\boxed{\neg (b \wedge c) \to \neg a}$

   $\neg 0 \to \neg 1$

   $= 1 \to 0$

   $= 0$

   v) $(b \wedge t) \to (a \vee r)$

   $? \to 1$

   $= 1$

   vi) $(a \vee r) \to (b \wedge t)$

   $1 \to ?$

   we dont know the answer

2. Construct a truth table for the statement $\neg(a \vee \neg b) \to \neg a$.

a	b	$\neg b$	$a \vee \neg b$	$\neg(a \vee \neg b)$	$\neg a$	$\neg(a \vee \neg b) \to \neg a$
0	0	1	1	0	1	1
0	1	0	0	1	1	1
1	0	1	1	0	0	1
1	1	0	1	0	0	1

3. Negate the statement $(a \vee \neg b) \to \neg c$ and simplify the result.

$\neg [(a \vee \neg b) \to \neg c]$

$\neg [\neg (a \vee \neg b) \vee \neg c]$ No NAME

$\neg\neg (a \vee \neg b) \wedge \neg\neg c$ DeMorgan's

$(a \vee \neg b) \wedge c$ Double Negation

4. Negate the sentence: If $\underbrace{a = b}_{x}$ and $\underbrace{c + d > 50}_{y}$ then $\underbrace{\text{this graph is bipartite}}_{z}$.

$\neg [(x \wedge y) \to z]$

$\neg [\neg (x \wedge y) \vee z]$ No NAME

$\neg\neg (x \wedge y) \wedge \neg z$ DeMorgan's

$x \wedge y \wedge \neg z$ Double Neg.

answer:

a=b and c+d>50 and this graph is NOT bipartite

Reason — "given" / Rule of logic Line # / Rule of inference Line #s

5. Consider the following argument and provide reason for each step

given
$\neg b$
$(\neg a \vee c) \rightarrow t$
$\neg a \vee n$
$\neg b \rightarrow \neg t$

$\therefore n \vee k$

Not "given" This is your target

	STEPS	REASON
1)	$\neg b$	given
2)	$\neg b \rightarrow \neg t$	given
3)	$\neg t$	M.P. 1,2
4)	$(\neg a \vee c) \rightarrow t$	given
5)	$\neg(\neg a \vee c)$	M.T. 3,4
6)	$a \wedge \neg c$	DeMorgan's 5
7)	a	C.S. 6
8)	$\neg a \vee n$	given
9)	n	D.S. 7,8
10)	$\therefore n \vee k$	D.A. 9

6. Use the rules of inference to show that the following arguments are valid. Provide the rule for each step.

a) $\neg t$
$\neg s$
$a \to t$

$\therefore \neg(a \lor s)$

$\neg a \land \neg s$
* this could be the second last line

STEPS	REASON
1) $\neg t$	given
2) $a \to t$	given
3) $\neg a$	M.T. 1,2
4) $\neg s$	given
5) $\neg a \land \neg s$	R.C. 3,4
6) $\neg(a \lor s)$	DeMorgans 5

b) $x \to (y \to z)$
x
$\neg y \to \neg x$

$\therefore z$

* every premise or a line can be used more than once if needed

STEPS	REASON
1) x	given
2) $x \to (y \to z)$	given
3) $y \to z$	M.P. 1,2
4) $\neg y \to \neg x$	given
5) y	M.T. 1,4
6) z	M.P. 3,4

c) $\neg s$
$p \to (q \to r)$
$t \to q$
$p \lor s$

$\therefore \neg r \to \neg t$

second last line could be
$t \to r$ (contraposit.)
$r \lor \neg t$ (no name)

STEPS	REASON
1) $\neg s$	given
2) $p \lor s$	given
3) p	D.S. 1,2
4) $p \to (q \to r)$	given
5) $q \to r$	M.P. 3,4
6) $t \to q$	given
7) $t \to r$	L.S. 5,6
8) $\neg r \to \neg t$	Contrapositive 7

d) $a \wedge b$
$\neg x$
$a \to (r \wedge b)$
$(r \vee n) \to (x \vee p)$

$\therefore p$

not needed, optional

watch for D.A. !

STEPS	REASON
1) $a \wedge b$	given
2) a	C.S. 1
3) b	C.S. 1
4) $a \to (r \wedge b)$	given
5) $r \wedge b$	M.P. 2,4
6) r	C.S. 5
7) $r \vee n$	D.A. 6
8) $(r \vee n) \to (x \vee p)$	given
9) $x \vee p$	M.P. 7,8
10) $\neg x$	given
11) p	D.S. 9,10

7. [redacted] prove the following argument:

$p \to q$
$(q \wedge r) \to s$
r

$\therefore p \to s$

STEPS	REASON
1) $(q \wedge r) \to s$	given
2) $\neg(q \wedge r) \vee s$	NONAME 1
3) $\neg q \vee \neg r \vee s$	DeMorgan 2
4) r	given
5) $\neg q \vee s$	D.S. 3,4
6) $q \to s$	NONAME 5
7) $p \to q$	given
8) $p \to s$	H.S. 6,7

NOTE $(q \wedge r) \to s$

$q \to s$

Not C.S. !
also not valid

$s \to (q \wedge r)$

$\therefore s \to q$

* not C.S. !

1) $s \to (q \wedge r)$ given
2) $\neg s \vee (q \wedge r)$ NONAME 1
3) $(\neg s \vee q) \wedge (\neg s \vee r)$ Distributive 2
4) $\neg s \vee q$ C.S 3
5) $s \to q$ NONAME 4

* this is C.S.

* observe this

If all of the problems can't be done in the lab period, students can work on them on their own.

No Name identity: $p \to q \Leftrightarrow \neg p \vee q$

Contrapositive: $p \to q \Leftrightarrow \neg q \to \neg p$

Rule of Inference	Related Logical Implication	Name of Rule
1) $\begin{array}{l} p \\ \underline{p \to q} \\ \therefore q \end{array}$	$[p \wedge (p \to q)] \to q$	Rule of Detachment (Modus Ponens)
2) $\begin{array}{l} p \to q \\ \underline{q \to r} \\ \therefore p \to r \end{array}$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$	Law of the Syllogism
3) $\begin{array}{l} p \to q \\ \underline{\neg q} \\ \therefore \neg p \end{array}$	$[(p \to q) \wedge \neg q] \to \neg p$	Modus Tollens
4) $\begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array}$		Rule of Conjunction
5) $\begin{array}{l} p \vee q \\ \underline{\neg p} \\ \therefore q \end{array}$	$[(p \vee q) \wedge \neg p] \to q$	Rule of Disjunctive Syllogism
6) $\begin{array}{l} \underline{\neg p \to F_0} \\ \therefore p \end{array}$	$(\neg p \to F_0) \to p$	Rule of Contradiction
7) $\begin{array}{l} \underline{p \wedge q} \\ \therefore p \end{array}$	$(p \wedge q) \to p$	Rule of Conjunctive Simplification
8) $\begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array}$	$p \to p \vee q$	Rule of Disjunctive Amplification
9) $\begin{array}{l} p \wedge q \\ \underline{p \to (q \to r)} \\ \therefore r \end{array}$	$[(p \wedge q) \wedge [p \to (q \to r)]] \to r$	Rule of Conditional Proof
10) $\begin{array}{l} p \to r \\ \underline{q \to r} \\ \therefore (p \vee q) \to r \end{array}$	$[(p \to r) \wedge (q \to r)] \to [(p \vee q) \to r]$	Rule for Proof by Cases
11) $\begin{array}{l} p \to q \\ r \to s \\ \underline{p \vee r} \\ \therefore q \vee s \end{array}$	$[(p \to q) \wedge (r \to s) \wedge (p \vee r)] \to (q \vee s)$	Rule of the Constructive Dilemma
12) $\begin{array}{l} p \to q \\ r \to s \\ \underline{\neg q \vee \neg s} \\ \therefore \neg p \vee \neg r \end{array}$	$[(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)] \to (\neg p \vee \neg r)$	Rule of the Destructive Dilemma